Sparse Attack with Meta-Learning

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***Abstract*—Black-box attacks pose a significant challenge due to the restricted access to target model information, hindering the generation of impactful adversarial samples. This paper introduces a method that combines sparse attacks and meta-learning to alleviate the issue of low success rates in black-box attacks. SAM leverages the knowledge transfer capabilities inherent in meta-learning to augment the transferability of adversarial samples. The method integrates meta-learning with gradient-based attack techniques, effectively transforming the approach into a white-box attack. By aggregating multiple sampled models, SAM enhances the stability of adversarial samples. During meta-testing, simulated black-box attacks help mitigate gradient discrepancies across diverse models, consequently enhancing transferability. To further improve sparsity and preserve transferability, SAM incorporates a projection strategy that selectively sparsifies global adversarial perturbations. Experimental evaluations conducted on two image datasets substantiate SAM's superiority in terms of both sparsity and attack success rate. Ablation experiments confirm the effectiveness of integrating meta-learning into the proposed method. SAM extends the applicability of generated adversarial samples, advancing the domain of adversarial attacks in scenarios with limited target model information. The proposed approach exhibits promise in enhancing the success rate of attacks while preserving sparsity, contributing to the broader understanding of black-box attacks and their implications.**

***Index Terms*—Sparse Attack, Meta Learning, Black-Box Attacks**

I. Introduction

In practical adversarial scenarios, attackers often encounter situations wherein they lack access to crucial information about the target model. Consider online image classification platforms as illustrative examples, where their functionality is constrained to providing labels based solely on user-inputted images. When facing a "black box" attack, where comprehensive details about the target model are unavailable, the challenge becomes more formidable. This is attributed to the inherent impossibility of leveraging specific information, such as the architecture and parameters of the target model, in devising attack strategies. Despite the increased difficulty associated with black box attacks compared to their white box counterparts, delving into these problems holds significant value. The rationale behind this pursuit lies in the broader applicability inherent to black box attacks, warranting dedicated research efforts.

In the realm of adversarial attacks, black box attacks manifest in two distinct forms: transfer-based and score-based black box attacks. The former predominantly targets a designated model by exploiting the transferability inherent in adversarial examples. In this context, transferability denotes the ability to successfully attack other models with disparate architectures and parameters while the adversarial examples originate from a single source model. However, it is essential to note that sparse attacks exhibit a significant reliance on the complexity of the architecture and gradient of the white box model. Consequently, adversarial examples generated through sparse attacks tend to overfit the white box model. This overfitting, in turn, imparts a weakened transferability, resulting in a declined success rate when employed in black box attack scenarios.

To effectively tackle the aforementioned challenges and expand the applicability of generated adversarial samples, our proposed approach introduces a sparse attack method grounded in meta-learning principles. This method is designed to produce adversarial samples characterized by optimal sparsity and enhanced transferability. Meta-learning facilitates the acquisition of knowledge and experience from analogous tasks, thereby empowering the model with the capacity to learn efficiently and generalize effectively across multiple tasks. Leveraging the inherent learning-to-learn capability and knowledge transfer potential of meta-learning, our methodology seeks to elevate the transferability of adversarial samples within the context of sparse attacks. This strategic integration of meta-learning principles serves to amplify the effectiveness of knowledge transfer, thereby contributing to the improved performance of adversarial sample generation in sparse attack scenarios.

The proposed sparse attack methodology expounded in this paper is delineated through a bifurcation into two distinct stages. During the initial phase, we employ a synthesis of

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the MAML algorithm architecture, rooted in meta-learning principles, and gradient-based attack methodologies renowned for their robust transferability. This amalgamation facilitates the generation of adversarial samples characterized by global perturbations. Subsequently, in the second stage, leveraging the acknowledgment that the adversarial perturbation crafted in the preceding phase is inherently global in nature, we introduce a projection strategy. This strategic intervention is pivotal, as it endeavors to induce sparsity in the global perturbation, culminating in the creation of final adversarial samples characterized by dual merits of notable sparsity and commendable transferability.

II. Related work

A. Sparse Attack

In this section, we delve into the extant body of work concerning sparse adversarial attacks. Unlike dense attacks, the crux of the sparse attack lies in discerning the optimal perturbation positions. Several methodologies have been introduced to address this challenge. JSMA [1] and its extensions have been advanced to identify the most influential pixels for model decision by leveraging saliency maps. CornerSearch [2] has proposed a heuristic strategy, navigating through all pixels and selecting a subset for perturbation. Sparse-RS [3] advocates a random search strategy for effective attack.

SparseFool formulates the problem as an ℓ_1 constraint problem, presenting a geometry-inspired sparse attack method. PGD0 [2] endeavors to project the adversarial noise generated by PGD [4] onto the $\ell 0$-ball, thus achieving an ℓ_0-norm version of PGD. ℓ_1-APGD [5] transforms the ℓ_0 norm constraint into the intersection of the ℓ_1-ball and the $[0,1]^d$ box constraints. Subsequently, it derives the correct steepest descent step for this intersection, motivating an adaptive sparsity in the chosen descent direction. The selection and adaptation of these diverse strategies contribute to the evolution of sparse adversarial attack methodologies.

B. Meta-learning

Meta-learning, also known as 'learning to learn,' enhances a network's adaptability and generalization by leveraging prior knowledge. This methodology proves particularly valuable in overcoming challenges posed by few-shot learning scenarios, where limited training data is available. The key lies in strategically transferring knowledge gained from analogous tasks to bolster multitask generalization capabilities. Meta-learning achieves this by automatically incorporating insights from other tasks, even when confronted with minimal training data.

The meta-learning process unfolds in two stages: meta-training and meta-testing. During meta-training, the method is trained on a group of tasks rather than individual ones. In the subsequent meta-testing stage, the method adeptly learns new tasks with limited training data. Notably, tasks in each stage are independent, featuring distinct support and query sets.

Two prevalent meta-learning architectures, Model-Agnostic Meta-Learning (MAML) and Reptile, address optimization concerns. This paper focuses on MAML, a model-agnostic technique applicable to a broad range of problems based on gradient learning. MAML's key principle involves initializing model parameters for robust generalization, fine-tuning them with minimal data for new tasks.

The MAML training workflow includes initializing network parameters (θ), updating support set parameters via gradient descent during meta-training, and computing query set loss. Aggregating losses across meta-training tasks, calculating the gradient, and updating parameter θ follow. This dual-cycle structure refines local optimum parameters in similar tasks through the interior cycle, while the exterior cycle systematically addresses losses in each task, updating the initialized parameter θ.

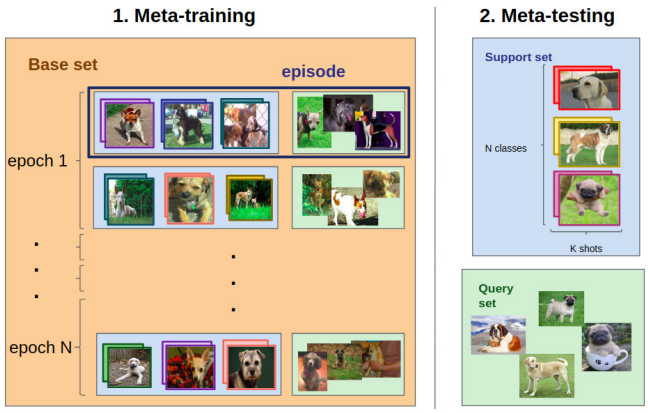


Fig. 1. Meta-learning

III. Method

In this section, we present a novel sparse adversarial attack leveraging meta-learning principles. The proposed method entails a two-fold process for generating sparse adversarial samples with enhanced transferability. The initial phase involves the integration of the meta-learning architecture Model-Agnostic Meta-Learning (MAML) to augment the efficacy of attack methods rooted in gradient-based approaches. By incorporating MAML, we aim to bolster the transferability of adversarial samples across diverse models. Subsequently, in the second phase, we employ a sparsification technique on the perturbations of adversarial samples, utilizing global perturbations. This dual-pronged approach seeks to concurrently imbue the adversarial samples with sparsity and optimize their transferability. The confluence of these methodologies results in the creation of adversarial samples that exhibit both sparse characteristics and heightened transferability, contributing to the advancement of adversarial attack techniques.

A. The Transferability Enhancing Method Based on Meta-Learing

In the succinct exposition on meta-learning, it becomes evident that meta-learning possesses the capacity for "learning to learn," enabling it to leverage the learning processes from

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analogous tasks. This unique attribute empowers meta-learning to augment a learner's generalization capability in novel tasks through knowledge transfer. Leveraging the cross-domain learning proficiency inherent in meta-learning, we introduce an approach that integrates the Model-Agnostic Meta-Learning (MAML) architecture within the meta-learning framework. Furthermore, we propose an adversarial attack method based on gradient manipulation, aiming to minimize the gradient disparities between adversarial samples in white-box and black-box scenarios through the utilization of MAML. The objective of this methodology is to bolster the transferability of adversarial samples across different domains. Subsequently, a detailed examination of the operationalization of this method is presented to elucidate its efficacy in enhancing the resilience of models against adversarial attacks.

Initially, we construct a model set comprising N white box models, each characterized by known parameters and architecture. These models serve as potential alternatives for conducting black box attacks. Subsequently, for each task within the meta-learning framework, we randomly select n models from this set, subject them to attacks, and generate corresponding adversarial samples, collectively constituting a task. This process is iteratively repeated across multiple tasks to yield the final set of adversarial samples.

In the context of a specific task, the n white box models are partitioned into two groups, mirroring the support set and query set in meta-learning. During the meta-training stage, a white box attack is executed, while a simulated black box attack is performed in the meta-testing stage. To illustrate, let N models be denoted as $M_1, M_2, \ldots, M_N$ within the set. From this set, we randomly sample n models, represented as $M_{a_1}, M_{a_2}, \ldots, M_{a_n}$ to constitute a task. The first $n-1$ models serve as the support set during the meta-training stage, while the nth model acts as the query set during the meta-testing stage.

Given that the final adversarial samples are generated through multiple tasks, each task can utilize the adversarial samples x_i generated in the prior task iteratively. Specifically, the initial input in every task is denoted as $x_{i,0}$. This iterative approach ensures the cohesive integration of adversarial samples across tasks, contributing to a robust and comprehensive evaluation of the proposed methodology.

Meta-training constitutes a pivotal stage in our approach, involving the utilization of the initial $n-1$ models denoted as $M_{a_1}, M_{a_2}, \cdots, M_{a_{n-1}}$. Within this phase, a collective white box attack strategy is adopted, wherein the focus extends beyond individual model vulnerabilities to encompass the holistic consideration of all $n-1$ models. This departure from a model-centric approach aims to enhance the overall efficacy of the adversarial attack. Concretely, the concurrent targeting of these $n-1$ models necessitates the computation of the cross-entropy loss function, defined as:

$$\mathcal{L}_{a_s}\left(x_{i,j}\right) = -\mathbb{1}_y \cdot \log\left(\operatorname{softmax}\left(l_{a_s}\left(x_{i,j}\right)\right)\right) \tag{1}$$

where y represents the authentic label of the sample, $\mathbb{1}_y$ denotes the one-hot encoding corresponding to this label, and $l_{a_s}\left(x_{i,j}\right)$ signifies the output logits for the input $x_{i,j}$ within the purview of model M_{a_s}. Following this computation, we proceed to evaluate the gradient of the current sample through backpropagation, subsequently undertaking a weighted average of the gradients from each model. This comprehensive meta-training strategy aligns with our objective of fortifying the robustness of the ensemble against adversarial manipulations.

$$\begin{aligned} \nabla_{x_{i,j}}^{a_s} &= \frac{\partial \mathcal{L}_{a_s}\left(x_{i,j}\right)}{\partial x_{i,j}} \\ \nabla_{x_{i,j}} &= \sum_{s=1}^{n-1} w_s \nabla_{x_{i,j}}^{a_s} \end{aligned} \tag{2}$$

where $\nabla_{x_{i,j}}^{a_s}$ means gradient of model M_{a_s} with current input sample $x_{i,j}$; $\nabla_{x_{i,j}}$ means holistic gradient calculated by weighted average merging; and w_s means the weight of each model, sufficient that $w_s \geq 0$, $\sum_{s=1}^{n-1} w_s = 1$. Through the equation listed above, we holistically considered the gradient to current sample of every model so that the adversarial sample will not overfit any simple model. Then we update the adversarial sample alongside the gradient increasing direction by maximizing the lost function:

$$x_{i,j+1} = x_{i,j} + \alpha \cdot \operatorname{sign}\left(\nabla_{x_{i,j}}\right) \tag{3}$$

where α is the perturbation magnitude of each step of iteration. In the meta-training stage, we can go multiple iteration updates, so the subscript j means current iteration times in this stage.

Meta-Testing: During this phase, we employ the final sampled model denoted as M_{a_n} to emulate a black-box attack scenario. Although we possess knowledge regarding the architecture and parameters of the model, it is crucial to underscore that this emulation does not constitute an authentic black-box attack; rather, it serves as a simulated approximation. This strategic choice is deliberate, aiming to minimize dissimilarities in gradient information across various models, thereby enhancing the transferability of adversarial samples. The rationale behind this simulation lies in fostering a more controlled environment for evaluating and refining the robustness of the model under consideration. Use the sample $x_{i,K}$ generated in the last one turn of meta-testing stage as this time's input, where K is the iteration times of this stage. Initially, we compute the cross-entropy loss function value of M_{a_n} with respect to the input $x_{i,K}$:

$$\mathcal{L}_{a_n}\left(x_{i,K}\right) = -\mathbb{1}_y \cdot \log\left(\operatorname{softmax}\left(l_{a_n}\left(x_{i,K}\right)\right)\right) \tag{4}$$

Subsequently, we calculate the derivative $\nabla_{x_{i,K}}^{a_n}$ with respect to $x_{i,K}$. In order to augment the transferability of adversarial samples, we directly incorporate the perturbation obtained during the meta-testing stage into x_i, generated in the preceding iteration:

$$x_{i+1} = x_i + \beta \cdot \operatorname{sign}\left(\nabla_{x_{i,K}}^{a_n}\right) \tag{5}$$

β denotes the single-step perturbation from the meta-testing stage.

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B. Global Perturbation Sparsification

Building upon the aforementioned procedures, our generated adversarial samples have exhibited commendable transferability. However, it is noteworthy that the perturbation applied to these samples encompasses the entire image, constituting a global perturbation. To endow the adversarial samples with both sparsity and enhanced transferability, it becomes imperative to introduce a mechanism for sparsifying the global perturbation. In this subsection, we will delineate the methodology employed for global perturbation sparsification.

For an original image sample denoted as $x \in [0,1]^{d\times 3}$, when a global perturbation δ is applied, the resulting adversarial sample is represented as x^{adv}. Subsequently, global perturbation sparsification is performed by projecting it onto the ℓ_0 spherical space, expressed as:

$$P(x) = \left\{ z \in \mathbb{R}^{d\times 3} \mid \sum_{i=1}^{d} \max_{j=1,2,3} \mathbb{1}_{\left|z_{ij}-x_{ij}\right|>0} \leq k,\ -\epsilon \leq z_{ij} - x_{ij} \leq \epsilon \right\} \quad (6)$$

where z represents the sparsified sparse adversarial sample, with k denoting the sparsity, signifying the maximum permissible number of pixel points with perturbation. ϵ represents the maximum perturbation amplitude on each individual pixel, while i and j represent the pixel coordinates. The indicator function $\mathbb{1}$ is utilized, with a value of 1 when the expression's value is less than 0. For ease of convenience, the projection problem can be reformulated as the following minimization problem:

$$\begin{aligned} &\min_{z\in\mathbb{R}^{d\times 3}} \sum_{i=1}^{d}\sum_{j=1}^{3} \left(x_{ij}^{adv} - z_{ij}\right)^2 \\ &\text{s. th. } -\epsilon \leq z_{ij} - x_{ij} \leq \epsilon, \quad i = 1,\ldots,d, j = 1,\ldots,3 \\ &\qquad \sum_{i=1}^{d} \max_{j=1,2,3} \mathbb{1}_{|z_{ij}-x_{ij}|>0} > 0 \leq k \end{aligned} \quad (7)$$

In order to concurrently adhere to the constraints of sparsity and perturbation amplitude, the sparsely constructed adversarial samples through projection should exhibit minimal divergence from the globally perturbed adversarial samples. This alignment ensures that, to the greatest possible extent, the information encapsulated within the adversarial samples prior to projection is preserved. Initially, we will set aside the challenge posed by the sparsity constraint and focus on resolving the pixel values post-projection for each individual pixel, as depicted below:

$$\begin{aligned} &\min_{z\in\mathbb{R}^{3}} \sum_{i=1}^{d}\sum_{j=1}^{3} \left(x_{ij}^{adv} - z_{ij}\right)^2 \\ &\text{s. th. } -\epsilon \leq z_{ij} - x_{ij} \leq \epsilon, \quad i = 1,\ldots,d, j = 1,\ldots,3 \end{aligned} \quad (8)$$

Normally those pixels added larger perturbation amplitude contribute more to generating adversarial samples so the value of pixels after projection is shown as:

$$z_{ij}^{*} = \max\left\{-\epsilon, \min\left\{x_{ij}^{adv} - x_{ij}, \epsilon\right\}\right\} \quad (9)$$

As each pixel point can be independently resolved, we can proceed by arranging them in descending order based on their respective contributions to the generation of samples:

$$\phi_i := \sum_{j=1}^{3} \left(x_{ij}^{adv} - x_{ij}\right)^2 - \sum_{j=1}^{3} \left(x_{ij}^{adv} - z_{ij}^{*}\right)^2 \quad (10)$$

Subsequently, to adhere to the sparsity constraint, we select the initial k pixels from the sorted list as the ultimate perturbed pixel value z_{ij}^{*}, while leaving the remaining pixels unchanged. This process yields the ultimate sparse adversarial samples z.

$$z_{ij} = \begin{cases} z_{ij}^{*} & \text{for } i = 1,\ldots,k, j = 1,\ldots,3 \\ x_{ij} & \text{else} \end{cases} \quad (11)$$

C. Algorithm

Algorithm 1 Sparse Attack with Meta-Learning(SAM)

Require: origin samples$x$;total task turns$T$;iteration turns in meta-training stage$K$and perturbation amplitudeα;meta-testing stage's perturbation amplitudeβ;quantity of perturbation pixels$k$;quantity of sampled models$n$;model's weightw_s

Ensure: adversarial samplesx^{adv}

initialize:$i \leftarrow 0$, $x_0 \leftarrow x$
while $i \leq T$ **do**
  randomly sample$n$models$M_{a_1}, M_{a_2}, \cdots, M_{a_n}$
  $x_{i,0} = x_i$
  meta-training stage
  $j = 0$
  while $j \leq K$ **do**
    $\mathcal{L}_{a_s}\left(x_{i,j}\right) = -\mathbb{1}_y \cdot \log\left(\text{softmax}\left(l_{a_s}\left(x_{i,j}\right)\right)\right)$
    $\nabla_{x_{i,j}} = \sum_{s=1}^{n-1} w_s \nabla_{x_{i,j}}^{a_s}$
    $x_{i,j+1} = x_{i,j} + \alpha \cdot \text{sign}\left(\nabla_{x_{i,j}}\right)$
    $j = j + 1$
  end while
  meta-testing stage
  $\mathcal{L}_{a_n}\left(x_{i,K}\right) = -\mathbb{1}_y \cdot \log\left(\text{softmax}\left(l_{a_n}\left(x_{i,K}\right)\right)\right)$
  $x_{i+1} = x_i + \beta \cdot \text{sign}\left(\nabla_{x_{i,K}}^{a_n}\right)$
  global perturbation sparsification, and quantity of permissioned perturbation pixels is k, as we mentioned in III-B section
  $x_{i+1} = P(x_{i+1})$
  $i = i + 1$
end while
$x^{adv} = x_T$
return adversarial samplesx^{adv}

In this section, we introduce the Sparse Attack with Meta-Learning (SAM) method, detailed in Algorithm 1. The SAM method generates adversarial samples through multiple iterations, involving three distinct stages in each iteration: meta-training, meta-testing, and global perturbation sparsification.

In every iteration, a set of n models is randomly sampled to formulate the meta-learning task for that iteration. During the meta-testing stage, the first $n-1$ sampled models are merged,

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TABLE I
MODELS IN THE SET. THE FIRST 6 ARE WHITE BOX MODELS AND THE LAST 6 ARE BLACK BOX MODELS IN THIS EXPERIMENT.

	ID	ImageNet	CIFAR-10
White Box Model	1	VGG19 [6]	VGG19 [6]
	2	VGG16 [6]	VGG16 [6]
	3	ResNet101 [7]	ResNet101 [7]
	4	ResNet152 [7]	ResNet152 [7]
	5	DenseNet201 [8]	DenseNet201 [8]
	6	DenseNet169 [8]	DenseNet169 [8]
Black Box Model	7	InceptionV3 [9]	ResNeXt50_32x4d [10]
	8	ResNet50 [7]	ResNet50 [7]
	9	DenseNet161 [8]	DenseNet161 [8]
	10	GoogLeNet [11]	GoogLeNet [11]
	11	MobileNetV2 [12]	MobileNetV2 [12]
	12	MnasNet1_0 [13]	MnasNet1_0 [13]

weighted by their respective gradient values. Subsequently, adversarial samples are generated through multiple iterations using a method based on gradient values. In the meta-testing stage, the last sampled model is employed to update the adversarial samples. Finally, the perturbation of the adversarial samples is sparsified, resulting in the generation of the ultimate sparse adversarial samples for that iteration, characterized by transferability.

IV. EXPERIMENTS AND RESULTS

In this section, we present the experimental analysis conducted on two image classification datasets. Initially, we provide an overview of the datasets used in the experiment, followed by a discussion on the metrics employed, parameter configurations, and the chosen comparison method. Subsequently, the results obtained from the experiments on the two datasets are presented. Finally, we delve into the ablation experiment for this method, providing a detailed analysis of the outcomes.

A. Datasets

This experiment is conducted using two widely utilized public datasets: ImageNet [14] and CIFAR-10 [15].

The ImageNet dataset, a cornerstone in computer vision, contains 14,197,122 images across 20,000 categories. ILSVRC2012, a prominent subset, features 1,000 categories with training, validation, and testing sets comprising 1,281,167, 50,000, and 100,000 images. To manage its scale, our experiment randomly selects 1,000 images from ILSVRC2012, ensuring diverse category representation.

CIFAR-10, designed for common object classification, offers 60,000 32x32 RGB images in 10 classes. The training set holds 50,000 images, with an additional 10,000 in the testing set. Our experiment randomly samples 5,000 images, evenly distributed across the 10 categories.

B. Metric

In this experiment, we employ the Attack Success Rate (ASR) to assess the efficacy of adversarial samples. When the object classification model aligns with the model generating the adversarial samples, ASR represents the success rate of a white box attack. Conversely, when the object classification model differs from the adversarial sample's generating model, ASR signifies the success rate of a black box attack, indicating the transferability of adversarial samples.

To evaluate the sparsity of adversarial samples, we employ the average ℓ_0 norm of adversarial perturbation as the metric. This norm represents the number of pixels that have undergone perturbation in the original samples. A smaller ℓ_0 value indicates enhanced sparsity in the adversarial samples.

C. Experiment setting and contrast method

Model: In our experiment, we employ a set of 6 white box models to generate adversarial samples. In each iteration of our proposed method, we randomly select more than one model from this set to constitute a meta-learning task. Additionally, for black box attacks, we incorporate 6 other models as target models. Details of both white box and black box models for ImageNet and CIFAR-10 datasets are presented in Table I.

For the ImageNet dataset, all 12 models utilize the official PyTorch-trained model, with each image input sized at $224 \times 224 \times 3$. In the case of CIFAR-10, all 12 models load PyTorch-trained parameters, undergo fine-tuning on CIFAR-10, and feature an input size of $32 \times 32 \times 3$.

Parameters: The experimental setup involves manipulating pixel values within the range of 0 to 255. To ensure comprehensive exploration, we consider two distinct perturbation amplitudes, specifically, 255 and 16. Normalization is applied to these values, resulting in normalized perturbation amplitudes of 1.0 and 0.062, respectively. Our investigation extends to comparative evaluations against SparseFool [16] and GreedyFool [17]. For SparseFool, a hyperparameter λ influences sparsity and success rate, prompting us to vary its values as 5, 10, and 15. In the case of GreedyFool, employing the C&W [18] loss function introduces the hyperparameter κ, governing attack strength. We systematically vary κ as 0, 5, and 10 in our experiments. Other parameters in SparseFool and GreedyFool remain consistent with their default settings. The parameters of our proposed method, outlined in this paper, are configured as follows: Total task turns $T = 40$; Iteration turns in the meta-training stage $K = 10$; The number of models randomly sampled in each turn $n = 5$; Model weight $w_s = 0.2$, evenly distributed across models; Perturbation amplitude in the meta-training stage $\alpha = \frac{\epsilon}{K}$; Perturbation

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TABLE II

WHEN LARGEST PERTURBATION AMPLITUDE $\epsilon = 255$, COMPARISON BETWEEN OUR METHOD AND OTHER METHODS OF ATTACK SUCCESS RATE ON IMAGENET AND ADVERSARIAL PERTURBATION'S AVERAGE ℓ_0NORM.IDS ARE THE SAME AS TABLE I.

Method	ℓ_0	white-box models						black-box models					
		1	2	3	4	5	6	7	8	9	10	11	12
SparseFool($\lambda = 5$)	238.8	100	23.5	7.58	6.86	7.17	8.19	5.51	6.93	5.71	15.5	10.2	10.0
SparseFool($\lambda = 10$)	355.8	100	31.9	10.2	7.90	9.32	10.9	8.04	10.2	7.83	17.4	11.6	15.2
SparseFool($\lambda = 15$)	452.1	100	36.7	13.1	10.6	11.3	11.9	10.4	13.2	10.1	20.7	15.4	17.6
GreedyFool($\kappa = 0$)	89.5	100	19.8	6.42	5.53	5.12	4.77	10.1	5.70	3.82	12.8	5.18	5.32
GreedyFool($\kappa = 5$)	210.4	100	39.9	16.2	12.7	12.3	13.2	19.5	15.7	11.2	25.1	12.1	12.6
GreedyFool($\kappa = 10$)	383.8	100	64.8	24.7	20.4	19.3	20.2	24.5	18.9	32.3	17.9	17.3	24.5
SAM(Ours)	400	68.5	62.2	73.1	73.3	64.3	65.7	**34.7**	**48.5**	**37.2**	**40.5**	**45.5**	**38.5**

TABLE III

WHEN LARGEST PERTURBATION AMPLITUDE $\epsilon = 16$, COMPARISON BETWEEN OUR METHOD AND OTHER METHODS OF ATTACK SUCCESS RATE ON IMAGENET AND ADVERSARIAL PERTURBATION'S AVERAGE ℓ_0NORM.IDS ARE THE SAME AS TABLE I.

Method	ℓ_0	white-box models						black-box models					
		1	2	3	4	5	6	7	8	9	10	11	12
SparseFool($\lambda = 5$)	803	100	12.8	3.12	3.40	5.23	5.36	2.31	2.80	3.91	9.59	6.64	2.36
SparseFool($\lambda = 10$)	2017	100	23.1	4.62	5.33	6.25	7.14	3.91	6.43	5.97	12.4	10.6	11.4
SparseFool($\lambda = 15$)	3705	100	36.2	6.67	6.82	8.53	9.65	5.53	9.14	8.26	15.5	15.2	14.7
GreedyFool($\kappa = 0$)	506	100	4.70	1.90	1.55	1.34	1.52	2.54	1.43	1.26	2.94	1.94	2.22
GreedyFool($\kappa = 5$)	1368	100	19.7	3.30	3.25	3.21	4.65	6.60	4.46	3.98	7.16	6.64	5.88
GreedyFool($\kappa = 10$)	2671	100	34.7	6.95	4.30	5.95	7.15	9.02	7.46	6.48	10.4	9.76	9.04
SAM(Ours)	3000	60.8	62.5	50.7	51.5	50.9	52.5	**19.5**	**23.2**	**20.2**	**25.3**	**24.3**	**24.6**

amplitude in the meta-testing stage $\beta = \epsilon$, where ϵ represents the largest perturbation amplitude, and K denotes the iteration turns in the meta-training stage.

D. Experiment Result Analysis

In this section, we delineate the comparative experimentation between our proposed method and other sparse attack techniques on both Imageset and CIFAR-10 datasets. Notably, our method requires information from multiple models, distinguishing it from SparseFool and GreedyFool, which operate efficiently with a single model. To facilitate a consistent comparison, we utilize VGG19 both as the source model and the white-box model for generating adversarial samples in the cases of SparseFool and GreedyFool. Consequently, in the table presented below, the success rate of VGG19 corresponds to the success rate of the white-box attack, while other success rates denote the success rate of the black-box attack. This approach ensures a standardized and meaningful evaluation of the attack methods under consideration.

1) result analysis of experiment on ImageSet: Tables II and III present the results of the non-targeted attack experiments on ImageSet, with the largest perturbation amplitude ϵ set to 255 and 16, respectively. In our method, the permissible perturbed pixel quantity, denoted as ℓ_0, is predetermined before experimentation. In contrast, SparseFool and GreedyFool employ a dynamic perturbed pixel quantity until the attack succeeds. To ensure a fair and informative comparative analysis, we adjust the perturbed pixel quantity for SparseFool and GreedyFool after obtaining their experimental results. Specifically, we set ℓ_0 at 3000 and 400 when ϵ is 255 and 16, respectively. This approach harmonizes the experimental conditions and facilitates a meaningful contrast between the attack methods under consideration.

Analyzing the results presented in Table II, it is evident that our method demonstrates a substantial increase in success rates compared to other methods when the perturbed pixel quantity is similar. Notably, when considering the InceptionV3 model, SparseFool and GreedyFool achieve success rates of 10.4% and 24.5%, respectively, in their attacks. In contrast, our method achieves a success rate of 34.7%, marking a significant improvement of 20.3% and 10.2%, respectively. This indicates that adversarial samples generated by our method exhibit superior transferability.

Furthermore, both SparseFool and GreedyFool achieve 100% success rates on the VGG19 white-box model, considerably higher than their success rates on black-box models. This suggests that their adversarial samples may be overfitting the white-box model. In contrast, our method achieves a success rate of 68.5% on VGG19, and similar success rates on all six white-box models. This highlights that our method's adversarial samples do not overfit any single model, demonstrating better transferability. This improvement can be attributed to our utilization of gradient information from multiple models in generating adversarial samples.

In the case of a perturbation amplitude of 16, where the

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permissible alteration amplitude on each pixel point of the image is reduced, generating adversarial samples becomes more challenging. As observed in Table III, compared to Table II, every ℓ_0 has increased, and every success rate has decreased. Despite this heightened difficulty, our method consistently outperforms SparseFool and GreedyFool in terms of success rates on black-box models. This indicates that adversarial samples generated by our method exhibit both favorable sparsity and remarkable transferability simultaneously, even in more challenging conditions.

2) CIFAR-10 Dataset's experiment result analysis: To assess our method's performance on different datasets, we conducted a similar experiment on CIFAR-10. The tables below present the results of non-targeted attacks by different methods on CIFAR-10, with perturbation amplitudes ϵ set to 255 and 16, respectively.

In comparison to the ImageNet dataset, CIFAR-10 has a smaller scale with only 10 categories, resulting in a lower level of attack difficulty. Analyzing the contrast in attack success rates presented in Table IV, when ϵ is set to 255, our method demonstrates significantly higher success rates on black-box models compared to SparseFool and GreedyFool. For instance, on the ResNeXt5_32x4d black-box model, our method achieves a success rate of 88.5%, while SparseFool and GreedyFool achieve only 44.0% and 17.8%, respectively. Notably, in Table IV, our method attains similar success rates on black-box models with unknown architectures and parameters as those on white-box models. This observation indicates that our method's adversarial samples are effective in attacking most models, showcasing remarkable transferability.

In CIFAR-10 experiments with ϵ set to 16, our method achieves a nearly 50% success rate in attacking all six black box models, while GreedyFool's success is limited to 10%. GreedyFool's reliance on a greedy search algorithm, driven by the C&W loss function's gradient information from a white box model, leads to overfitting and weak transferability of adversarial samples. In contrast, our approach employs random sampling and diverse model combinations during adversarial perturbation updates, enhancing transferability to unknown black box architectures. The holistic use of gradient information from multiple models prevents dependency on any single model, avoiding overfitting. In conclusion, our Sparse Attack with Meta-learning ensures sparsity and superior transferability compared to alternative methods.

3) Ablation Experiment Analysis: To substantiate the effectiveness of the Model-Agnostic Meta-Learning (MAML) approach employed in this paper, this section introduces an ablation experiment, providing insights into the impact of MAML on the experimental results.

The experimental results for non-targeted attacks on CIFAR are presented in Table VI. The SA method denotes an approach that solely relies on attacking based on gradient and global perturbation sparsification, excluding the meta-training and meta-learning stages proposed in this paper. Within the SA framework, only VGG19 is utilized as the white box model. Notably, when we maintain the perturbation amplitude at the same level, there is a significant increase in the success rate of attacking black box models, as evidenced by the data in the table. This observation substantiates the efficacy of the MAML architecture in enhancing the transferability of adversarial samples, as discussed in this paper.

V. Conclusion

Addressing the challenge of low success rates in black-box attacks, we present the Sparse Attack with Meta-learning method to enhance the transferability of adversarial samples. Inspired by meta-learning's knowledge transfer, we integrate it with a gradient-based attack method, combining multiple models for white-box attacks. In the meta-testing phase, we reduce gradient differences between models, boosting transferability. A projection strategy enhances sparsity and preserves transferability in adversarial samples. Comparative experiments on two image datasets demonstrate our method's excellence in both sparsity and attack success rates, with an ablation experiment confirming the effectiveness of the meta-learning approach.

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TABLE IV

WHEN LARGEST PERTURBATION AMPLITUDE $\epsilon = 255$, COMPARISON BETWEEN OUR METHOD AND OTHER METHODS OF ATTACK SUCCESS RATE ON CIFAR-10 AND ADVERSARIAL PERTURBATION'S AVERAGE ℓ_0NORM.IDS ARE THE SAME AS TABLE I.

Method	ℓ_0	white-box models						black-box models					
		1	2	3	4	5	6	7	8	9	10	11	12
SparseFool($\lambda = 5$)	33.2	100	27.7	27.6	20.3	19.4	25.7	29.3	23.1	26.7	23.2	25.6	26.8
SparseFool($\lambda = 10$)	49.9	100	44.3	32.9	28.2	29.1	31.8	34.7	34.6	35.7	36.5	34.4	39.4
SparseFool($\lambda = 15$)	79.2	100	50.9	38.0	37.8	40.2	38.3	44.0	43.4	45.8	45.7	39.5	46.9
GreedyFool($\kappa = 0$)	19.2	100	20.8	11.2	14.4	14.2	7.6	10.4	13.9	11.5	16.5	10.2	15.8
GreedyFool($\kappa = 5$)	35.2	100	33.1	15.8	20.9	20.7	10.8	14.1	19.4	16.9	22.5	14.5	22.0
GreedyFool($\kappa = 10$)	56.7	100	42.8	20.2	26.2	26.3	13.8	17.8	23.6	23.4	27.9	17.9	27.4
SAM(Ours)	50	89.9	87.7	90.0	92.8	93.6	93.3	**88.5**	**88.1**	**91.5**	**86.6**	**79.4**	**78.8**

TABLE V

WHEN LARGEST PERTURBATION AMPLITUDE $\epsilon = 16$, COMPARISON BETWEEN OUR METHOD AND OTHER METHODS OF ATTACK SUCCESS RATE ON CIFAR-10 AND ADVERSARIAL PERTURBATION'S AVERAGE ℓ_0NORM.IDS ARE THE SAME AS TABLE I.

Method	ℓ_0	white-box models						black-box models					
		1	2	3	4	5	6	7	8	9	10	11	12
SparseFool($\lambda = 5$)	192	100	20.7	23.9	14.1	13.7	21.3	27.0	18.5	22.0	16.7	24.2	24.0
SparseFool($\lambda = 10$)	383	100	32.2	25.6	18.2	18.2	23.9	28.8	21.1	24.7	20.7	26.7	30.6
SparseFool($\lambda = 15$)	556	100	36.1	26.1	21.7	19.4	25.2	29.0	22.5	26.2	22.8	27.6	33.0
GreedyFool($\kappa = 0$)	87	100	5.37	2.66	4.08	3.56	1.82	2.02	3.76	1.88	4.82	4.01	8.04
GreedyFool($\kappa = 5$)	165	100	11.3	4.29	6.46	6.26	3.03	3.32	5.62	2.85	7.09	6.76	12.6
GreedyFool($\kappa = 10$)	284	100	17.5	5.25	8.82	9.26	3.70	4.64	7.62	4.39	9.49	8.86	16.5
SAM(Ours)	300	66.4	63.4	58.5	62.3	61.0	63.7	**43.9**	**50.8**	**48.5**	**45.8**	**43.3**	**57.0**

TABLE VI

MAML ARCHITECTURE'S ABLATION EXPERIMENT RESULT ON CIFAR-10.IDS ARE THE SAME AS TABLE I.

Method	ℓ_0	white-box models						black-box models					
		1	2	3	4	5	6	7	8	9	10	11	12
SA($\epsilon = 255$)	50	91.8	81.6	71.4	74.7	75.2	79.3	67.2	64.3	68.2	73.8	67.3	69.8
SAM($\epsilon = 255$)	50	89.9	87.7	90.0	92.8	93.6	93.3	**88.5**	**88.1**	**91.5**	**86.6**	**79.4**	**78.8**
SA($\epsilon = 16$)	300	80.6	30.6	26.2	18.8	15.4	25.8	31.3	23.2	25.8	20.8	24.4	28.2
SAM($\epsilon = 16$)	300	66.4	63.4	58.5	62.3	61.0	63.7	**43.9**	**50.8**	**48.5**	**45.8**	**43.3**	**57.0**

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